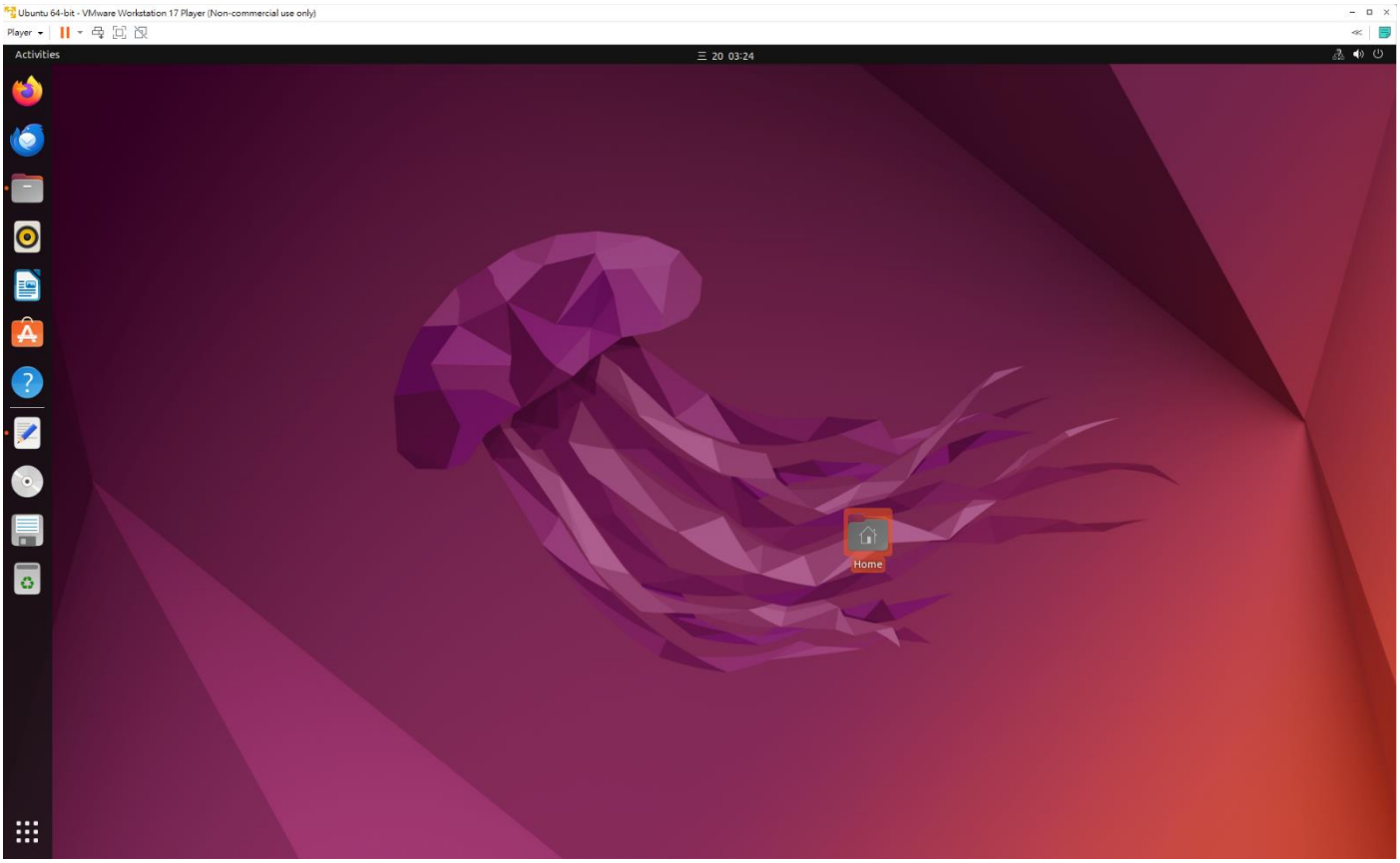
Embedded Operating System HW01 A111193

Part 1: Prepare Environment for future homework

1.

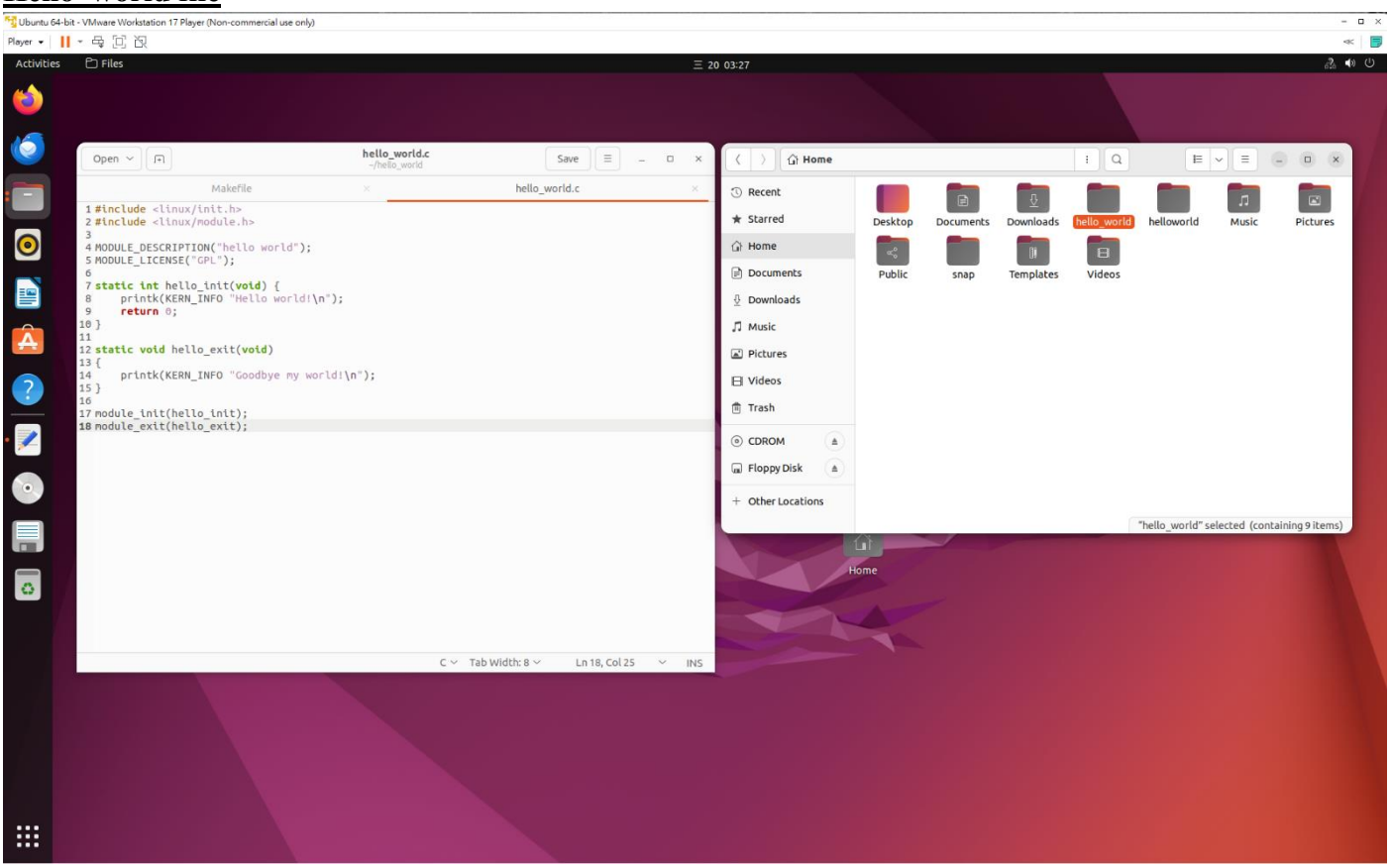
<u>Ubuntu Desktop</u>



Part 2: Learn how to write a kernel module

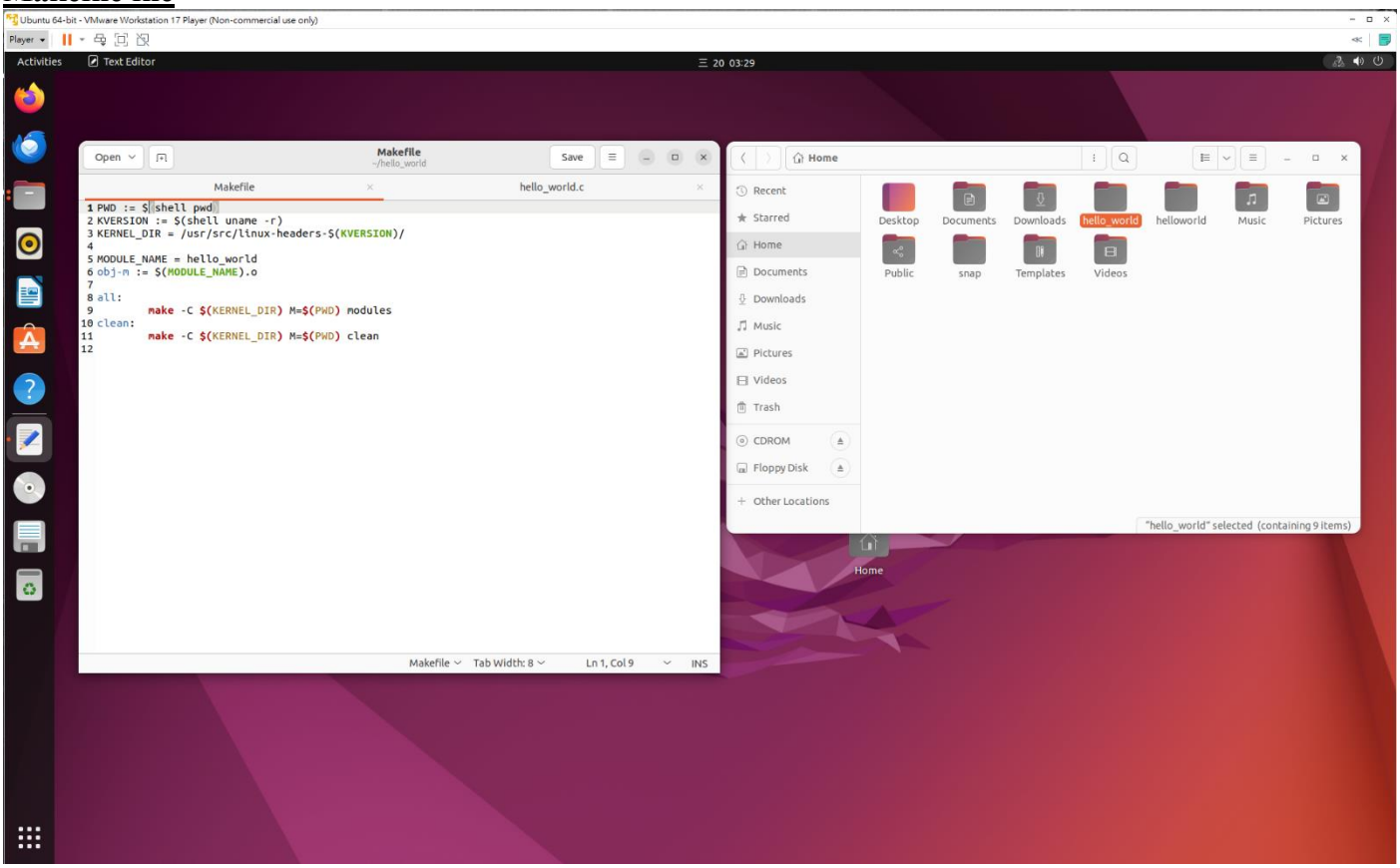
1.

<u>Hello_world file</u>



2.

Makefile file

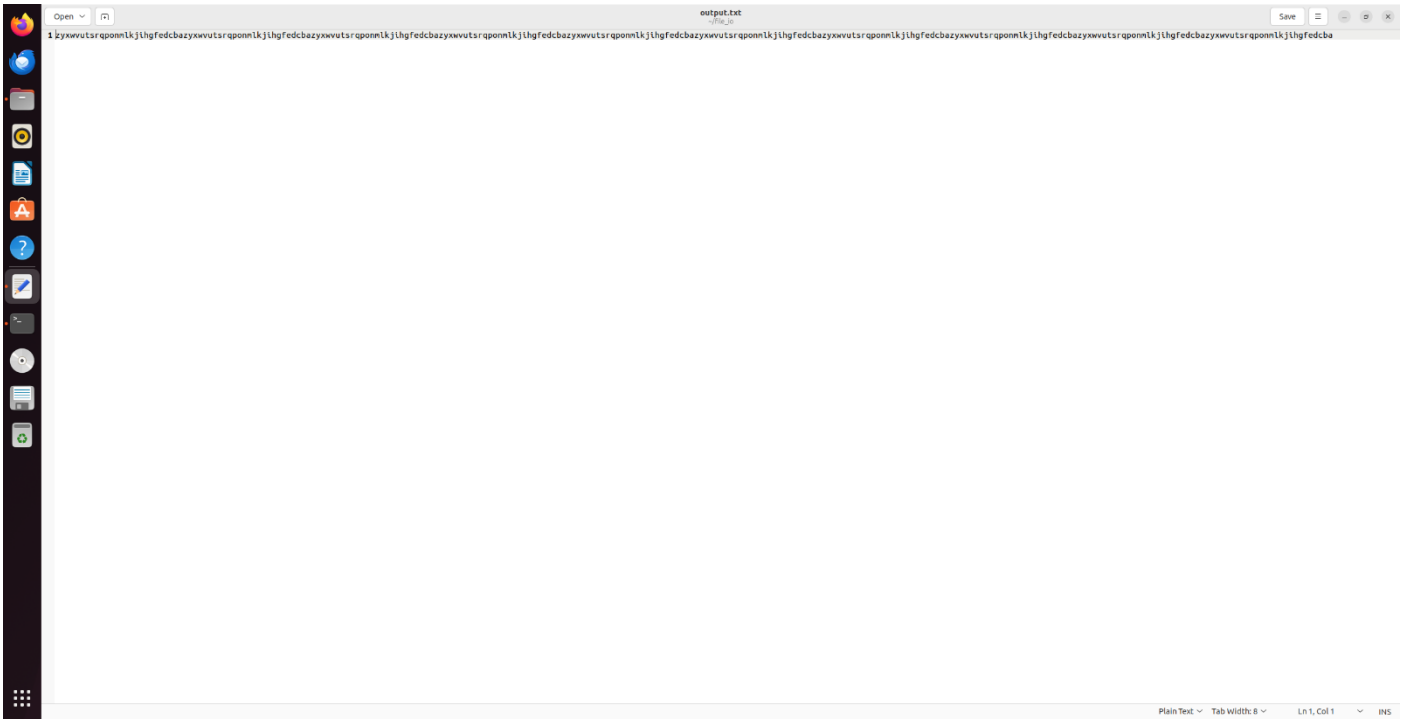


3.

Dmesg of hello_world

```
[22992.496948] usb 1-2.1: reset full-speed USB device number 4 using uhci_hcd
[27169.565357] e1000: ens33 NIC Link is Down
[27173.597775] e1000: ens33 NIC Link is Up 1000 Mbps Full Duplex, Flow Control: None
[27219.963652] e1000: ens33 NIC Link is Down
[27223.995226] e1000: ens33 NIC Link is Up 1000 Mbps Full Duplex, Flow Control: None
[69437.587928] floppy0: disk absent or changed during operation
[69437.587981] I/O error, dev fd0, sector 0 op 0x0:(READ) flags 0x80700 phys_seg 1 prio class 2
[69437.588056] floppy0: disk absent or changed during operation
[69437.588057] I/O error, dev fd0, sector 0 op 0x0:(READ) flags 0x0 phys_seg 1 prio class 2
[69437.588061] Buffer I/O error on dev fd0, logical block 0, async page read
[69437.588351] ext3: Unknown parameter 'utf8'
[69437.588365] ext2: Unknown parameter 'utf8'
[69437.588372] ext4: Unknown parameter 'utf8'
[69437.588380] squashfs: Unknown parameter 'utf8'
[77933.789174] hello_world: loading out-of-tree module taints kernel.
[77933.789182] hello_world: module verification failed: signature and/or required key missing - tainting kernel
[77933.804939] Hello world!
[77999.918332] Goodbye my world!
user@user-virtual-machine:~/hello_world$
```

Part 3:Contribution

1.

<u>Output.txt file</u>